Return to Education in the U.S. with College Proximity Instrument: A Deep Learning Approach with Satellite Imagery

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Introduction

Estimating the returns to education- how much additional income individuals earn from an additional year of schooling is a fundamental question in economics. A seminal contribution by Card (1995) uses college proximity whether an individual lives close to a college as an instrument for education. This instrument is plausibly exogenous, as individuals' locations at age 14 are determined prior to their college attendance decisions. Cards estimates therefore identify the return to education for a marginal population those who attend college only because they live nearby. However, this instrument is constructed as a binary indicator of whether a college exists within a county, which may be a coarse measure of proximity. This paper extends Cards approach by introducing a continuous measure of college proximity and incorporating spatial features such as forests, rivers, and road networks.

Research Question:

How sensitive are IV estimates of returns to education to the inclusion of detailed geographic controls?

Data and Methods

Dataset:

- American Community Survey Public-use Microdata Area (ACS PUMA)
  - Education attainment, wage, age, gender, race
- IPEDS
  - College Location (longitude, latitude)
- Microsoft Planetary Computer
  - Satellite Images

Sample: U.S. Working-age individuals who live in the same home since 14

Model: Two-stage Least Squares (TSLS)

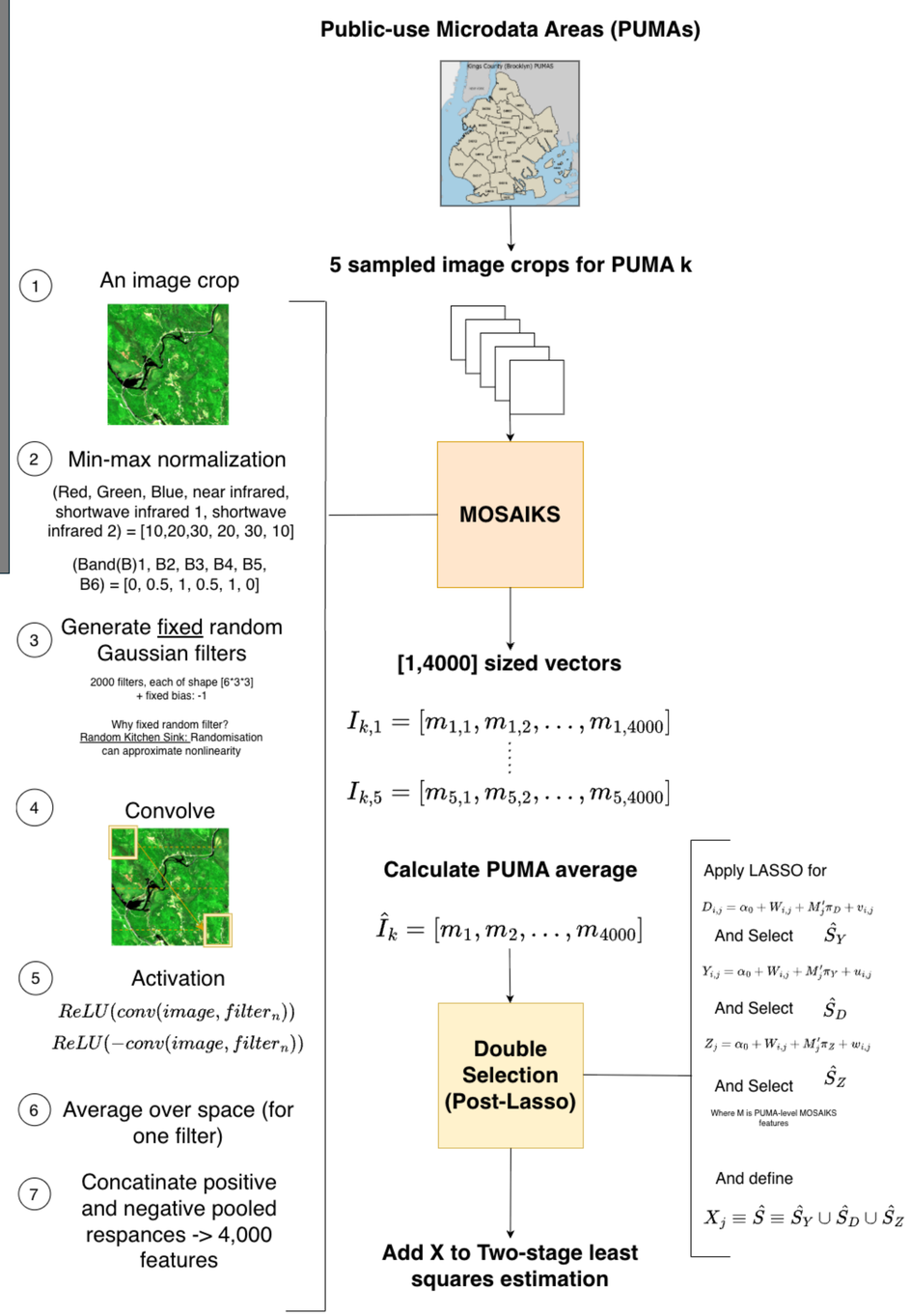
$$Y_{i,j} = \alpha_0 + \theta D_{i,j} + W'_{i,j}\alpha_1 + X'_j\alpha_2 + \epsilon_{i,j}$$

$$D_{i,j} = \beta_0 + \delta Z_j + W'_{i,j}\beta_1 + X'_j\beta_2 + \zeta_{i,j}$$

Where:
Y - Log Wage
D - Years of Education
Z - Distance to the Nearest College
W - Demographics
X - Selected Image Features

For individual i and PUMA j

Image Feature Creation (X)



Regression Results

	OLS	IV baseline	
	(1)	(2)	(3)
Education	0.079***	0.190***	0.207***
	(0.001)	(0.015)	(0.015)
Demographics	Yes	No	Yes
PUMA FEs	Yes	No	No
Image Features	No	No	No
N. of Image Features			
Observations	148,770	147,451	147,451
First-stage F-stat		153.23	158.31

IV-PUMA FEs		IV-img	
(4)	(5)	(6)	(7)
0.143***	0.162***	0.168***	0.180***
(0.044)	(0.043)	(0.042)	(0.041)
No	Yes	No	Yes
Yes	Yes	No	No
No	No	Yes	Yes
		187	195
147,451	147,451	146,638	146,638
42.66	47.57	37.72	39.32

Notes: Standard errors in parentheses are clustered at the PUMA level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Results

- The baseline TSLS estimates yield returns of 19.0% and 20.7% without and with demographic controls, respectively, with strong first-stage F-statistics.
- When PUMA fixed effects are included, the estimated returns decline to 14.3% and 16.2%, suggesting that geographic and local labor factors at the PUMA level account for a substantial portion of the variation.
- After controlling for satellite-derived geographic features, the estimated returns are 16.8% and 18.0% without and with demographic controls, respectively.
- This indicates that image features absorb part of the geographic variation approximately 2 percentage points, but not as fully as PUMA fixed effects.